

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester B. Tech (ME) Examination

March-April 2018

ME 305.01/ME 305 Manufacturing Processes-II**Date: 05.04.2018, Thursday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- (i) Give specification criteria for engine lathe. **[03]**
- (ii) A 150 mm long 25 mm diameter rod is reduced to 23 mm diameter in a single pass straight turning. If the cutting speed is 15 m/min and feed rate is 225 mm/min, find: **[04]**
 - i. Material removal rate
 - ii. Cutting time.

Q - 2. Answer any three questions.**[12]**

- (i) What are the lathe accessories? Give the name and use of various lathe accessories.
- (ii) Explain the construction and operation of back geared head stock of a lathe.
- (iii) Explain the following lathe operations with neat figure:
 1. Parting
 2. Grooving
- (iv) State and explain different methods of holding Drill tool in a Drilling machine Spindle.
- (v) Calculate the time required to drill a hole 20 mm diameter in a gray cast iron workpiece 70 mm thick using a high speed steel drill. The cutting speed and the feed rate for the operation may be assumed to be 0.60 m/s and 0.6 mm per revolution of the drill respectively.

Q - 3 Answer any four questions.**[16]**

- (i) What is meant by Straddle milling and Gang milling operations? Explain.
- (ii) Sketch & compare Climb milling and Conventional milling.
- (iii) List main parts of shaper machine and state their function.
- (iv) Explain the terms: (1) Truing and (2) Dressing.
- (v) Write short note about Broaching Machine.
- (vi) Give specifications of grinding wheel.

SECTION – II**Q - 4 Answer the following.**

- (i) Differentiate hot metal working process and cold metal working process. **[05]**
- (ii) What do you mean by trimming operation? **[02]**
- (iii) Explain the terms: billet, bloom, sheet, plate. **[04]**

Q - 5 Answer any three questions. [12]

- (i) Explain board hammer with neat sketch.
- (ii) Explain forward extrusion process with neat sketch.
- (iii) Give any two examples of rolling stand arrangements with neat sketch.
- (iv) Which kind of defects can be generated in rolling process? Discuss reasons behind it.
- (v) Determine the engineering strain, true strain, and reduction for (1) a bar which is doubled in length (2) a bar which is halved in length.

Q - 6 Answer any three questions. [12]

- (i) What is forging? Enlist the advantages and drawbacks of forging process compared to other manufacturing processes.
- (ii) Explain wire drawing process with neat sketch.
- (iii) What is the use of mandrel in extrusion process? Explain with figure.
- (iv) Determine the die and punch size for punching a circular disc of 35 mm diameter from a C20 steel whose thickness is 2.8 mm. allowable shear stress = 294 MPa.
